

Tasks for the next week :

OpenFOAM:

Main goal for next week:

- Freeze the thermal-hydraulic assumptions and start a representative cooling model.

Tasks

1. Define the thermal-hydraulic design basis:
 - reactor thermal power: 125 MWth
 - inlet temperature
 - outlet temperature
 - operating pressure
 - coolant density
 - mass flow rate estimate

Ask Samira for:

- hottest assembly power
- axial power shape
- peaking factors
2. Build the first OpenFOAM geometry:
 - representative hot channel or single assembly
 - coolant region
 - heated wall/fuel rod simplification
3. Prepare boundary conditions:
 - inlet temperature
 - inlet velocity or mass flow
 - outlet pressure
 - heat flux or volumetric heat source
4. Run one simple steady-state case, even if coarse.

Save:

- mesh screenshot
- temperature contour
- velocity contour
- pressure drop result

Deliverable by end of week:

- A first OpenFOAM steady-state hot-channel result with assumptions table.

OpenMC :

Main focus: results generation and OpenMC report package.

Tasks:

1. Freeze OpenMC Baseline v1.
2. Export all 1.2 figures and tables.
3. Finish MTC sweep.
4. Finish void coefficient sweep.
5. Create final DTC/MTC/void coefficient table.
6. Start all-rods-in control rod worth case.
7. Prepare OpenMC digital appendix folder and README.
8. Prepare fuel/material design tables.
9. Check material compositions and densities.
10. Prepare Gd tuning table from Samira's sweep results.
11. Help define control rod absorber material and geometry.
12. Run or help process sensitivity cases:
 - enrichment perturbation
 - Gd perturbation
 - water density perturbation
13. Draft Section 8.3 Fuel and Material Design.

If I have more time :

14. Start **shutdown margin** case
15. Start simplified **depletion test**. Track :
 - k_{eff}
 - U-235
 - U-238
 - Pu-239
 - Pu-240
 - Pu-241
 - Gd-155
 - Gd-157
16. Sensitivity mini-study

3S safety / I&C / FOM

Main goal for next week:

- Build the safety and I&C framework that connects everyone's technical results.

Tasks

1. Create a safety criteria table:
2. Draft I&C architecture:
 - neutron detectors
 - temperature sensors
 - pressure sensors
 - flow sensors
 - reactor protection system
 - DCS
 - digital twin monitoring layer
3. Draw first I&C block diagram.
4. Define trip signals:
 - high neutron flux
 - high outlet temperature
 - low flow
 - high pressure
 - low steam generator level
5. Start one event tree:
 - station blackout, or
 - loss of heat sink
6. Prepare FOM update template:
 - input from OpenMC
 - input from OpenFOAM
 - input from TES/SOE
 - input from layout/economics

Deliverable by end of week:

- A safety criteria table + I&C block diagram + one event tree draft.

Energy conversion / TES / SOE

Main goal for next week:

- Make the non-electrical application part numerical and consistent.

Tasks

1. Freeze the energy balance
2. Prepare a flow diagram:
 - reactor → steam generator → turbine → electricity
 - steam/heat extraction → TES district heating
 - off-peak operation → SOE hydrogen production
3. Update TES sizing:
 - storage energy target
 - charge/discharge temperature
 - mass of zeolite or storage material
 - district heating output
4. Update SOE hydrogen calculation:
 - input power
 - operating hours
 - kWh/kg H₂
 - kg H₂/day
 - kg H₂/year
5. Give Elbek approximate space/layout needs:
 - TES module area
 - SOE unit area
 - hydrogen storage area
6. Start simple economics:
 - electricity revenue estimate
 - heat revenue estimate
 - hydrogen revenue estimate
 - rough CAPEX/OPEX assumptions

Deliverable by end of week:

- An energy balance table + TES/SOE calculation table + flow diagram.

NPP layout / auxiliary systems

Main goal for next week:

- Create the first facility layout concept.

Tasks

1. Make a block-level facility layout with zones:
 - reactor building
 - containment / nuclear island
 - turbine building
 - control room
 - auxiliary building
 - spent fuel / waste area
 - TES, SOE
 - cooling water / heat rejection area
2. Draw major flow arrows:
 - main steam
 - feedwater
 - electrical output
 - district heat
 - hydrogen
 - waste/fuel handling
3. Prepare a building list (what buildings do we have)
4. Start auxiliary systems list:
 - ventilation
 - fire protection
 - radiation monitoring
 - emergency power
 - electrical distribution
 - fuel handling
 - waste handling
5. Ask Adilbek for piping/flow requirements.
6. Ask S.Alisher for TES/SOE space requirements.

Deliverable by end of week:

- A first 2D block layout + building/function table + auxiliary systems list.